# Beyza KAYA

**Address:** Dresden, GERMANY **Mobile:** +49 155 10649317
**E-mail:** h.kaya@hzdr.de **Website:** https://beyzoskaya.github.io/

## OBJECTIVE

- Doctoral research on measuring what large language models know, using ecological sampling and richness estimation against known reference universes, to build reliable methods for evaluating knowledge in foundation models.

## EDUCATION

**TU DRESDEN & CASUS, HZDR** - Dresden, GERMANY (2026 - Present)
**PhD Candidate – Helmholtz School for Integrated Data Science in Environmental and Life Sciences (IDEAS)**
**Thesis Topic:** How much does an LLM know? Sampling hidden knowledge like an ecologist
**Supervisors:** Prof. Simon Razniewski (ScaDS.AI, TU Dresden), Prof. Justin Calabrese (CASUS, HZDR)

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2024 - 2026)
**Graduate School of Engineering and Science - Master's in Artificial Intelligence –** Completed
**Honors:** 100% Performance Scholarship **Cumulative GPA:** 4.00/4.00

**ÖZYEĞİN UNIVERSITY** - Istanbul, TURKEY (2019 - 2024)
**Faculty of Engineering - Department of Industrial Engineering –** Completed in 3.5 years
**Department of Computer Science Engineering (Double Major)**
**Honors:** 50% Performance Scholarship **Cumulative GPA:** 3.38/4.00

**PAMUKKALE EGITIM VAKFI HIGH SCHOOL** - Denizli, TURKEY (2015 - 2019)

## PROJECTS & RESEARCH STUDIES

**TU DRESDEN & CASUS, HZDR** - Dresden, GERMANY (2026 - Present)
**Position:** Doctoral Researcher (IDEAS)
**Research Project:** Measuring hidden knowledge in LLMs with ecological sampling (In Progress)

- Treating each LLM answer as a sample and each distinct entity as a species, and using accumulation curves, singletons, doubletons and Chao estimators to measure how much of a domain a model knows.
- Built fixed reference universes from Wikidata (e.g., 5,295 living mammal species) and a reproducible elicitation and matching pipeline for four open models, with Wikidata lookup and a validated LLM judge.
- Comparing independent, batch and open-listing prompts, and testing richness estimators on simulated communities with known distributions.

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2025-2026)
**Position:** Research Assistant
**Research Project:** ProteinGLM — Multi-Task Learning with Protein Language Models (In Progress)

- Designing a generalized multi-task learning framework for protein sequence modeling using ProtBERT as a shared encoder to jointly predict multiple biologically relevant downstream tasks.
- Implementing parameter-efficient fine-tuning via LoRA to enable scalable adaptation of large protein language models across tasks with minimal task-specific parameters.
- Exploring semantic prompting and task-conditioning strategies to teach the shared model task-specific objectives while preserving shared representations.
- Source code available at: https://github.com/beyzoskaya/PGM-TL

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2024-2026)
**Position:** Research Assistant
**Research Paper:** "Temporal Expression Prediction by Integrating Genome Dynamics via Spatio-temporal GNNs" (Under Review)
**Authors:** Beyza Kaya, Emre Sefer

- Proposed STEPmr and STEPmi, two spatio-temporal graph neural network models to predict mRNA and miRNA expression dynamics over time.
- Integrated Hi-C–derived spatial gene interaction features with temporal expression profiles to model gene regulatory dependencies.
- Analyzed gene ontology and biological pathway enrichments to interpret model predictions, identifying gene groups related to signaling, transcriptional regulation, and structural organization as key contributors to model behavior.
- Source code available at: https://github.com/seferlab/temporalgene

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2024-2026)
**Position:** Research Assistant

**Research Paper:** "GAT-HiC Efficient Reconstruction of 3D Chromosome Structure via Graph Attention Neural Networks" (Accepted by IEEE Transactions on Computational Biology and Bioinformatics )

**Authors:** Beyza Kaya, Emre Sefer

- Developed GAT-HiC, a novel graph neural network model combining Node2vec and graph attention mechanisms to predict 3D chromsome structures.
- Designed specialized loss functions to enhance generalization tasks across diverse species and datasets.
- Validated model performance across three Hi-C interaction datasets, demonstrating generalizability compared to existing methods.
- Source code available at: https://github.com/beyzoskaya/GAT-HiC

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2023 - 2024)

**Position:** Senior Design Project Computer Science

- Designed a generative framework for producing high-quality and stylistically rare NFT images using Stable Diffusion guided by reinforcement learning.
- Defined a rarity-aware reward function and integrated it into a PPO-based reinforcement learning pipeline to promote rare visual attributes while preserving prompt fidelity.
- Balanced rarity maximization and visual coherence through KL-regularized policy updates, enabling controlled exploration of the generative space.
- Trained diffusion models on large-scale NFT image datasets and evaluated generation quality under both in-domain and out-of-domain prompts, demonstrating robustness to non-NFT textual inputs.
- Source code available at: https://github.com/seferlab/diffnftgen

**ADEL KALEMCILIK -** Istanbul, TURKEY (2022)

**Position:** Senior Design Project Industrial Engineering

- Collaborated with a multidisciplinary team of Industrial Engineering students, an academic supervisor, and industry engineers from ADEL Kalemcilik on a real-world manufacturing optimization project.
- Collected and analyzed production line data through on-site observations and process documentation to model the end-to-end fabrication workflow.
- Designed and implemented a discrete-event simulation model using Arena to identify bottlenecks and inefficiencies in the production line.
- Evaluated alternative process configurations and operational scenarios through simulation experiments.

## TEACHING & STUDENT MENTORSHIP

### Teaching Assistantships

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2024-2026)

**Position:** Teaching Assistantships

**CS 201 – Data Structures and Algorithms**

- Assisted weekly laboratory sessions by answering students' questions on data structures and algorithm implementation.
- Graded midterm and final examinations, ensuring consistency and fairness in evaluation.

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2024-2026)

**Position:** Teaching Assistantships

**CS 104 – Introduction to Programming (Python)**

- Led weekly interactive lab sessions, guiding students through hands-on programming exercises.
- Conducted live coding and problem-solving sessions, explaining core programming concepts and best practices.
- Supported students individually during labs to reinforce understanding of course material.

**ÖZYEĞİN UNIVERSITY -** Istanbul, TURKEY (2024)

**Position:** Teaching Assistantships

**CS 333 – Analysis of Algorithms**

- Graded homework assignments and term projects for undergraduate students.
- Evaluated algorithmic correctness, complexity analysis, and clarity of technical explanations.

### Undergraduate Project Mentorship

**Senior Design Project – DeepAllergen: Deep Learning–Based Identification of Allergenic Proteins (In Progress)**

**Position:** Research Mentor

- Supervised undergraduate Computer Science students on a senior design project focused on allergen prediction from protein sequences.
- Guided students in problem formulation, dataset preparation, model architecture design, and experimental evaluation.
- Provided mentorship on deep learning methods for biological sequence modeling and scientific reporting.

**Senior Design Project – Protein Allergen Classification via Graph-Based Modeling (In Progress)**

**Position:** Research Mentor

- Supervised undergraduate Computer Science students on a senior design project focused on graph-based approaches for protein allergen classification.
- Advised students on graph modeling of biological data, feature engineering, and model evaluation strategies.

## EXPERIENCE

**DELTAV SPACE TECHNOLOGIES** - Istanbul, TURKEY (07/2023 - 08/2023)

**Position:** Intern, Project Management

- Analysed risk accordance with standards
- Planned project with detailed excel project for automation
- Predicted GPS latitude-longitude over IMU/ GPS data via Tensorflow and Keras

**UP SCHOOL** - Istanbul, TURKEY (11/2022 - 12/2022)

**Position:** Intern, Social Media Organizer

- Shared daily posts, videos and announcements on social media
- This internship was part of the Young Guru Academy's initiatives

**SNI TECHNOLOGY** - Istanbul, TURKEY (06/2022 - 08/2022)

**Position:** Intern, Java Developer

- Observed reflections of security systems on e-archive billing applications with Spring Framework
- Used the Spring Framework for email verification and wrote register services

**POILABS** - Istanbul, TURKEY (6/2022 - 10/2022)

**Position:** Intern, Backend Developer

- Wrote newly defined JavaScript features for hardware to use in shopping centers
- Participated project meetings on the integration of the company device for new airport

## VOLUNTARY ACTIVITIES

**YOUNG GURU ACADEMY -** Istanbul, TURKEY (2019 - 2021)

**Position:** Project Leader

- Led “Develop Business Model" sessions for 5th & 6th grade students to support their development
- Participated in camps on leadership, teamwork, communication, problem-solving, decision-making.

**Position:** Science Activities Volunteer

- Organized sessions with elementary school students in village schools with no access to education.
- Utilized "Twin" science kits and conducted experiments in groups.

**Position:** Summit Volunteer

- Dealt with the speakers and attendees and took part in various preparations during the summit.
- Produced creative content and created videos for YGA’s Instagram and TikTok accounts

## SKILLS & CERTIFICATES

**COMPUTER SKILLS**

- MS Office (Word, Excel, Power Point)
- Java (Intermediate)
- C++ (Intermediate)/C (Intermediate)
- Spring Framework (Intermediate)
- Tensorflow (Intermediate)
- PyTorch (Intermediate)
- Python (Intermediate)

**LANGUAGE SKILLS**

- Written and oral fluency in English

**CERTIFICATES**

PyTorch for Deep Learning Bootcamp

## EXTRACURRICULAR ACTIVITIES

- Member, Ozyegin University Google Developer Club (2020 - 2023)
- Member, Ozyeğin University SAS Club (2019 - 2021)
- Member, Ozyeğin University Business Club (2019 - 2021)
- Member, Ozyeğin University Operations Research Club (2019 - 2021)
- Member, Ozyeğin University Dance Club (2019 - 2020)
- Volleyball: Denizli Belediyespor Club Volleyball Team player (2013 - 2019)

## PERSONAL INFORMATION

**Birth Date:** 06/06/2001

**Driver’s License:** B Class